Neural Networks and Deep Learning Lab 10

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Objective

The objective of this experiment is to understand the Transformer architecture and its core components such as self-attention, multi-head attention, and positional encoding. Additionally, the experiment focuses on implementing a sequence-to-sequence model for Hindi to English translation and analyzing its advantages over traditional sequence models like RNNs and LSTMs.

Methodology

Dataset Preparation

A Hindi-English parallel corpus was used for training the model. The dataset was preprocessed by converting all text to lowercase, removing noise, and applying tokenization. Separate vocabularies were created for the source and target languages. Special tokens such as <PAD>, <SOS>, <EOS>, and <UNK> were introduced to handle padding, sequence boundaries, and unknown words.

Each sentence was converted into a numerical sequence and padded to maintain uniform input length. The dataset was then split into training and testing sets.

Transformer Architecture

The Transformer architecture replaces recurrence with attention mechanisms, enabling efficient parallel computation. The central component is the scaled dot-product attention defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Multi-head attention allows the model to attend to different parts of the sequence simultaneously. Positional encoding is used to preserve word order information:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d}}\right), \quad PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

The architecture consists of stacked encoder and decoder layers, each incorporating attention mechanisms, feed-forward networks, residual connections, and layer normalization.

Training

The model was trained using cross-entropy loss and the Adam optimizer. Teacher forcing was applied during training to improve convergence. Training was conducted over multiple epochs, and performance was monitored using training and validation loss and accuracy.

Translation Task

The trained model takes Hindi sentences as input and generates English translations. The performance was evaluated using BLEU score and qualitative comparison of predicted outputs.

Observations

The Transformer model demonstrates strong capability in capturing long-range dependencies due to its attention mechanism. Unlike RNN-based models, it processes sequences in parallel, leading to faster training and improved gradient flow.

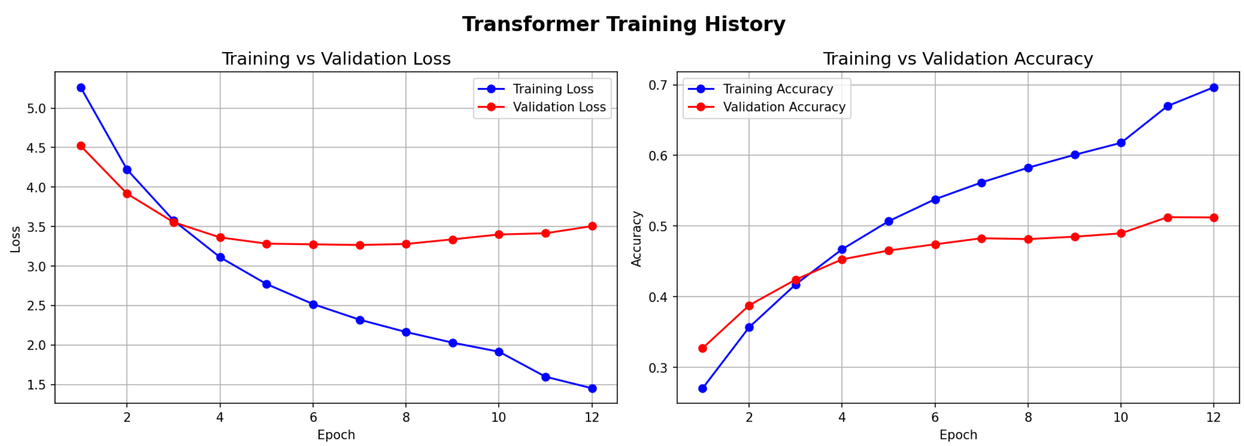

Figure 1: Training vs Validation Loss and Accuracy

From the training curves, it is observed that the training loss decreases steadily across epochs, indicating effective learning. However, the validation loss decreases initially and then stabilizes, with a slight increase in later epochs. This behavior suggests the onset of overfitting.

Similarly, training accuracy increases consistently and reaches approximately 70%, while validation accuracy plateaus around 50%. The gap between training and validation performance indicates that the model is learning training data patterns well but struggles to generalize to unseen data.

Prediction Analysis and BLEU Score

The model's performance was further evaluated using sample translations and BLEU score. The obtained BLEU score is:

$$\text{BLEU Score} = 0.1158$$

This low BLEU score indicates poor translation quality and limited generalization ability. Some example predictions are shown below:

- English: how are you
  Predicted:
  Observation: Correct translation
- English: What is your name?
  Predicted:
  Observation: Missing the word "", leading to incorrect meaning
- English: Who are you?
  Predicted:
  Observation: Grammatically incorrect and semantically incorrect

From these examples, it is evident that the model captures basic sentence patterns but struggles with grammar, word selection, and sentence structure.

The low BLEU score can be attributed to limited dataset size, insufficient training, and vocabulary constraints. Additionally, the observed overfitting further reduces the model's ability to generalize effectively.

Conclusion

Transformers effectively replace recurrence with attention mechanisms, enabling parallel processing and better handling of long-range dependencies. Positional encoding ensures that sequence order is preserved.

While Transformers provide significant advantages such as faster training and improved performance, they require large datasets and computational resources for optimal results.

Inference

The experiment demonstrates that Transformer models are powerful for machine translation tasks. However, achieving high-quality translations requires sufficient data, proper tuning, and careful handling of overfitting.